Igor, championships and paints

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

It would seem that with the start of the internship at Yandex, it's time to say goodbye to competitive programming. But it was not there — once more, Someone got involved in the creation of competitive programming championships...

— The same *Someone*

Igor recently started an internship at Yandex. He was given a lot of interesting stuff on his first day, including n jars, each with its own color of paint.

But since Igor doesn't know a thing about the fine arts, he had no idea to do with it all. 'Mathematics is art too, though!' he thought before deciding to do a little calculating. He assigned the numbers 1, 2, $\ldots$, n to the paints and then determined *paint mixing sum* as

$$\sum_{i=1}^{n}\sum_{j=1}^{n} i\,AND\,j$$

modulo 10^9+7, where $x\,AND\,y$ means "bitwise AND" of the numbers x and y.

Now Igor wants to know what *paint mixing sum* is equal to, but he's so busy with work that he doesn't have time to figure it out himself. So he asked you for help!

Input

The first line contains a single integer t $(1 \le t \le 10^3)$ — the number of testcases.

The only line of each testcase contains a single integer n $(1 \le n \le 10^{18})$ — number of paints.

Output

For each testcase, print answer modulo 10^9+7.

Example

standard input	standard output
5 1 2 8 1024 1000000000000000000	1 3 120 268174336 969095352

Note

Consider the first and the second testcases.

- In the first testcase,
  - $(1\,\&\,1) = 1$;

The sum equals 1. $1 \bmod (10^9+7) = 1$, so *paint mixing sum* equals 1.

- In the second testcase,
  - $(1 \,\&\, 1) = 1$;
  - $(1 \,\&\, 2) = 0$;
  - $(2 \,\&\, 1) = 0$;
  - $(2 \,\&\, 2) = 2$;

  The sum is $1+0+0+2=3$. $3 \bmod (10^9+7) = 3$, so *paint mixing sum* equals 3.